Understanding Transferable Urban Mobility Structure: Distance Decay Recovery and Interpretable Zero-Shot Prediction

Thinh Nguyen[1*]

[1*]Aarista Technologies, Ho Chi Minh City, Vietnam.

Corresponding author(s). E-mail(s): nqt900@gmail.com;

Abstract

Developing transferable mobility models for cities without origin-destination (OD) observations remains an open challenge. To address the difficulties and high costs of collecting individual mobility traces, which are increasingly restricted by strict privacy laws, recent research has shifted toward utilizing open data. However, existing frameworks still exhibit limited transferability. To bridge this gap, we propose the Separable Decomposed Framework for Zero-Shot Transferable Flows (SDF-ZTF), a decomposed mobility modeling approach that generates OD flows by integrating open-data-driven indicators with a production-constrained gravity model. By relying exclusively on globally available open datasets, including OpenStreetMap and WorldPop gridded population data, this methodology enables mobility estimation for previously unseen cities without requiring local OD observations, surveys, or target-city retraining. A key contribution of this work is a cross-city feature transferability analysis that identifies a compact set of stable urban indicators capable of generalizing across heterogeneous cities. The decomposed architecture not only improves interpretability by isolating the contribution of each mobility mechanism but also provides a practical pathway for diagnosing model limitations and guiding future improvements. Experimental results demonstrate that our approach achieves performance comparable to state-of-the-art deep learning models under fully zero-shot transfer settings while maintaining transparency and policy relevance.

Keywords: Urban mobility, origin-destination flows, gravity models, interpretability, transfer learning, zero-shot transfer, open-data modeling

1 Introduction

Predicting origin-destination (OD) flows is central to transportation planning and urban management (Barbosa et al. 2018; González et al. 2008; Song et al. 2010). Yet traditional OD data collection relies on household travel surveys that are expensive, slow to update, and unavailable in many cities—especially in developing regions.

Two modeling paradigms have emerged to fill these gaps. Classical gravity and radiation models (Wilson 1971; Simini et al. 2012; Stouffer 1940) offer interpretable, physically motivated predictions but need local calibration data. Deep learning models such as DeepGravity (Simini et al. 2021) and graph neural networks (Rong et al. 2023; Wang et al. 2021) achieve higher accuracy but act as black boxes and require target-city OD data for training. In short, interpretable models lack transferability, and transferable neural models lack transparency.

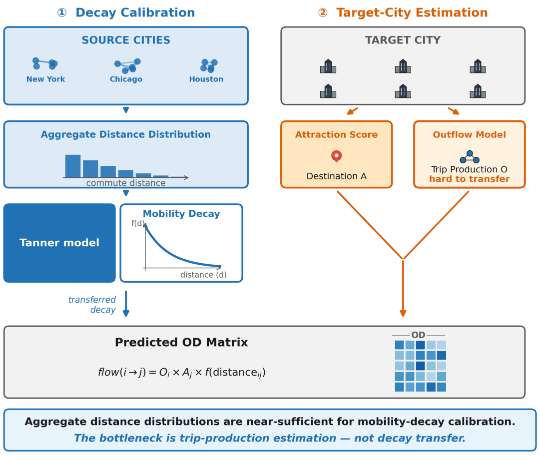

Fig. 1 SDF-ZTF framework: Three independent components (origin production O_i from features, distance decay $f(d)$ from aggregate data, destination attraction A_j from features)

Meanwhile, aggregate trip-length distributions—compiled from mobile devices and location-sharing platforms—are increasingly available even where full OD matrices are not. These summaries do not expose individual trajectories and can be shared under strict anonymization policies. A natural question is whether such aggregate data can calibrate the distance decay function for a new city without requiring individual OD records, and which component of spatial interaction (production, attraction, or decay) matters most for prediction quality.

We investigate three research questions:

1. Can aggregate travel-distance distributions calibrate the distance decay function without individual OD trajectories?
2. Which component—production, attraction, or distance decay—contributes most to flow prediction accuracy?
3. Can a decomposed, interpretable model match neural baselines under zero-shot cross-city transfer using only open data?

We propose the **Separable Decomposed Framework for Zero-Shot Transferable Flows** (**SDF-ZTF**), which factorizes OD flows into three independent components: origin production, distance decay, and destination attraction. The framework uses only globally available open data—OpenStreetMap Atwal et al. (2025) and WorldPop Tatem (2017)—making it deployable in any city without local surveys or retraining. Fig. 2 compares this zero-shot paradigm with the standard locally dependent approach.

This study makes three contributions:

- **Decay Recovery**: Aggregate distance distributions are sufficient to recover decay parameters without individual trajectories.
- **Component Ranking**: Distance decay is the dominant mechanism, accounting for 75.7% of explainable variance in cross-city transfer.
- **Zero-Shot Parity**: The decomposed framework matches DeepGravity under zero-shot conditions while remaining interpretable and using only open data.

2 Related Work

2.1 Classical and Structured Mobility Models

Gravity models (Wilson 1971; Tanner 1961) and radiation models (Simini et al. 2012; Alis et al. 2021) form the foundation of spatial interaction modeling. Gravity models use physical analogies to relate flows to zone mass and distance, while radiation models (Stouffer 1940) frame destination choice as a probabilistic process over intervening opportunities. Scaling-law studies (Brockmann et al. 2006; González et al. 2008) and predictability analyses (Song et al. 2010) have confirmed that travel distances follow robust statistical patterns. These models are interpretable and parsimonious, but gravity models need local OD data for parameter estimation, and radiation models underperform in cities where commuters routinely bypass closer opportunities.

2.2 Deep Learning for OD Prediction

DeepGravity (Simini et al. 2021) maps urban features to flow rates using multi-layer perceptrons. Follow-up work has applied graph neural networks (Rong et al. 2023), spatial-temporal networks (Wang et al. 2021), and OSM-based commuting models (Atwal et al. 2025). These approaches capture non-linear relationships and improve accuracy when training data is abundant, but they operate as black boxes, generalize poorly

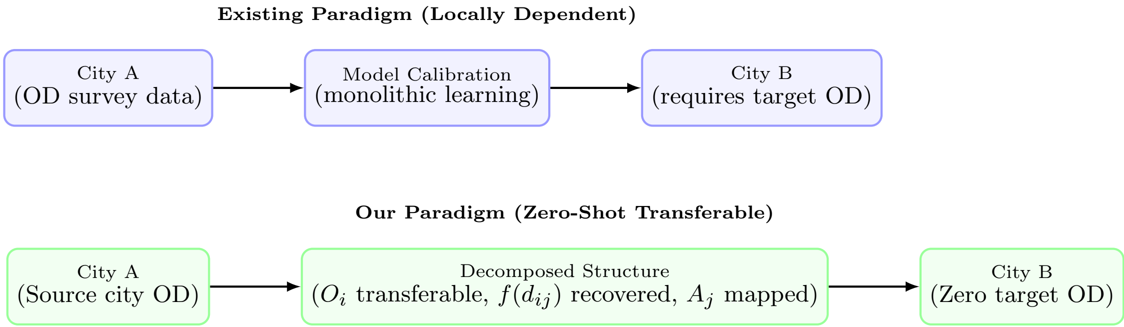

Fig. 2 Conceptual comparison of the existing locally dependent modeling paradigm versus our zero-shot transferable paradigm. Under the existing paradigm, models require target-city OD observations to retrain or calibrate parameters. In our paradigm, we learn transferable origin production structures from source City A, recover decay from target aggregate data, and map attraction from local open features in target City B, enabling zero-shot transfer without target-city OD surveys.

out of distribution, and require large amounts of target-city OD data.

2.3 Aggregate Mobility Data for Decay Calibration

Location-aware mobile devices now generate population-level mobility signals (Barbosa et al. 2018; González et al. 2008) that can be aggregated into trip-length distributions—histograms of how far people travel—without exposing individual OD pairs (Pappalardo et al. 2023). Several studies have shown that such distributions capture the spatial friction of urban travel and can fit distance decay functions (Brockmann et al. 2006; Lenormand et al. 2012, 2016). However, no prior work has used aggregate distributions specifically as a calibration signal for zero-shot cross-city transfer.

2.4 Cross-City Transfer Learning

Cross-city transfer learning aims to predict mobility in cities without OD surveys. Existing methods fall into three groups:

1. **Similarity-based**: RegionTrans (Wang et al. 2018) matches source and target zones by spatial similarity; CrossTReS (Jin et al. 2022) re-weights source regions via meta-learning.
2. **Representation learning**: Geospatial embeddings and GeoAI foundation models (Mai et al. 2022; Mendieta et al. 2023) learn general-purpose urban features.
3. **Domain adaptation**: STAN (Wang et al. 2022) and meta-spatiotemporal networks (Yao et al. 2019) align source-trained models to target environments.

All three groups still require some target-city observations, calibration, or fine-tuning.

2.5 Explainable and Transferable Urban Representations

SHAP (Lundberg and Lee 2017) is widely used in spatial modeling to explain feature importances, but almost always within a single city. Whether SHAP-based feature rankings are stable across cities—and can therefore guide transferable feature selection—has not been systematically studied.

2.6 Research Gap and Positioning

Table 1 compares existing model classes with our proposed SDF-ZTF.

No existing approach combines interpretable decomposition, open-data-only features, zero-shot cross-city transfer, and systematic transferable feature discovery. This study addresses four gaps:

Gap 1 (Calibration): Gravity models require local OD surveys for decay calibration.
Gap 2 (Interpretability): Neural models sacrifice transparency for accuracy.
Gap 3 (Data-dependency): Transfer frameworks still need target-city OD data.
Gap 4 (Feature Generalizability): No systematic method identifies which spatial features transfer across cities.

Table 1 Comparative analysis of human mobility models across structural and transferability dimensions. "Calibration-Free Decay" indicates whether the model recovers the distance decay function without requiring individual OD pair observations from target cities.

Model Class	Decomposed Structure	Cross-City Transfer	Zero-Shot Target	Interpretability	Calibration-Free Decay
Gravity (Wilson 1971)	✓	×	×	✓	×
Radiation (Simini et al. 2012)	×	✓	✓	✓	✓
DeepGravity (Simini et al. 2021)	×	Limited	×	×	×
GNN OD (Rong et al. 2023)	×	Limited	×	×	×
RegionTrans (Wang et al. 2018)	Partial	✓	×	×	×
SDF-ZTF (Ours)	✓	✓	✓	✓	✓

3 Methodological Design

3.1 Decomposition Hypothesis

We hypothesize that OD flows can be decomposed into three separable, city-independent components:

$$T_{ij} = O_i \cdot f(d_{ij}) \cdot A_j \tag{1}$$

where O_i represents origin production capacity, $f(d_{ij})$ represents distance decay, and A_j captures destination attraction. We hypothesize that: (1) each component is approximately independent, (2) the underlying mechanisms can be learned from spatial descriptors of the built environment, and (3) the learned origin production relationship transfers directly to unseen target cities, while decay parameters are recovered from target-city aggregate trip-length distributions, and attraction is computed from locally available open data. This decomposition hypothesis is directly tested by evaluating which of these three components exerts the dominant influence on predictive accuracy. Section 5 evaluates and validates this hypothesis empirically.

3.2 Experimental Framework for Investigating Transferable Mobility Structure

The conceptual pipeline of the proposed Separable Decomposed Framework for Zero-Shot Transferable Flows (SDF-ZTF) is illustrated in Fig. 3.

3.3 Distance Decay Function

We use the Tanner multiplicative deterrence function (Tanner 1961), which combines power-law and exponential terms:

$$f(d_{ij};\, \beta, \gamma, \delta) = (d_{ij} + \delta)^{-\gamma} \cdot \exp(-\beta\, d_{ij}) \tag{2}$$

where $\beta > 0$ is the exponential decay rate governing long-range trip attenuation, $\gamma \geq 0$ is the power-law exponent modeling short-to-medium range trip distribution, and $\delta > 0$ is the adaptive geometric anchor.

The offset δ prevents singularity at $d_{ij} = 0$ and adapts to zone size:

$$\delta = 0.5 \times \bar{R}_{\text{zone}}, \qquad \bar{R}_{\text{zone}} = \frac{1}{N} \sum_{i=1}^{N} \sqrt{\frac{\text{Area}_i}{\pi}} \tag{3}$$

where $\bar{R}_{\text{zone}}$ is the mean zone radius. Parameters (β, γ) are estimated via MLE on the aggregate distance-bin distribution. Calibration and optimization details are in the Appendix. Because only binned trip-length histograms are used—not individual OD records—this procedure is compatible with strict data-governance regulations (e.g., GDPR).

3.4 Origin Outflow Component (O_i)

The trip production $O_i = \sum_j T_{ij}$ is the total trips originating from zone i. We predict O_i with a GBDT regressor trained on 22 features selected via cross-city SHAP stability (Section 4) from a 30-dimensional candidate pool of open spatial data. Hyperparameters are in Appendix B.

3.5 Destination Attraction Component (A_j)

Destination attraction A_j measures how attractive zone j is to travelers. We compute it with a training-free heuristic: $\hat{A}_j \propto \frac{1}{2}\left(\text{minmax}(P_j) + \text{minmax}(\text{POI}_j)\right)$. Section 5 validates this choice by showing that a trained GBDT attraction model performs comparably,

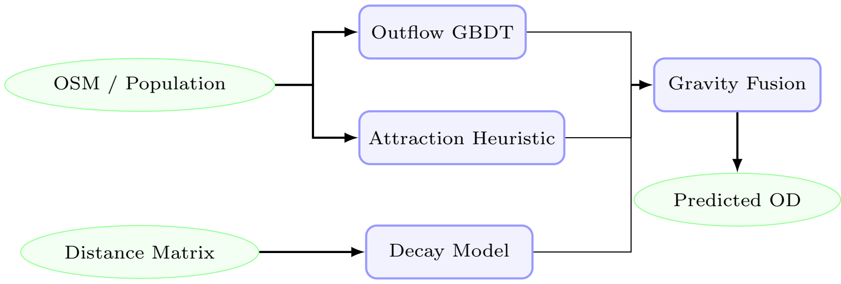

Fig. 3 Pipeline of the SDF-ZTF framework. Globally available spatial features are used to predict trip production (O_i) via a GBDT model and to compute destination attraction (A_j) exclusively via a training-free min-max heuristic (a GBDT attraction model is trained only during ablation validation to prove that this simple heuristic is sufficient, avoiding unnecessary complexity for cross-city transfer). Distance decay is estimated separately from aggregate distributions. The components are combined via multiplicative gravity fusion to predict zero-shot flow matrices (T_{ij}).

confirming that the heuristic is sufficient for cross-city transfer.

3.6 Algorithmic Flow

The complete pipeline for training and zero-shot deployment is formalized in Algorithm 1.

4 Data and Transferable Urban Representation

4.1 Design Principle: Open-Data-Only

All features in SDF-ZTF come from globally available open data rather than proprietary cellular records or household surveys. This makes the framework deployable in any city. We intentionally exclude local employment databases (e.g., LODES), which lack global equivalents, and instead proxy employment activity through commercial and industrial POI densities from OpenStreetMap.

4.2 Data Sources

We use three data sources. Their roles across framework components are shown in Table 2.

4.2.1 (1) Origin-Destination (OD) Matrix - T_{ij}

Seasonal OD flows between census tracts are extracted from Meta's Movement Distribution Maps (Meta AI 2026) for 50 US metropolitan areas. The OD matrix serves only as training labels for source cities and as evaluation ground truth; it is never used as a predictive feature.

4.2.2 (2) OSM-Derived Features

We extract POIs and road features using the Overpass API from OpenStreetMap snapshots (Atwal et al. 2025). POI counts are calculated across 16 land-use categories (e.g., retail, industrial, education, health).

4.2.3 (3) Population Data

Population counts come from the original mobility dataset but can be replaced with WorldPop (Tatem 2017) gridded estimates (100 m resolution) in cities lacking census data.

4.3 Transferable Urban Features

We use a two-stage procedure—feature engineering followed by stability selection—to identify spatial patterns that generalize across cities. The feature taxonomy is shown in Fig. 4.

4.3.1 Candidate Feature Set (30 dimensions)

We initially construct 30 candidate features per zone. This candidate space comprises 23 scale and density variables (zone area, road length, residential population, total POI counts, and 16 POI categories) and 7 city-wide Z-score context features. The Z-score context features are defined as:

$$z_{ip} = \frac{x_{ip} - \mu_p^{(c)}}{\sigma_p^{(c)}} \tag{4}$$

where $\mu_p^{(c)}$ and $\sigma_p^{(c)}$ represent the mean and standard deviation of feature p across all zones in city c. These context features normalize absolute

Algorithm 1: SDF-ZTF Training and Zero-Shot Prediction

Input: Source cities data $\{\mathbf{X}^{(c)}, \mathbf{T}^{(c)}\}_{c=1}^{C}$; Target city spatial features $\mathbf{X}^{(t)}$
Output: Predicted Target City OD Matrix $\hat{\mathbf{T}}^{(t)}$
1. **Decay Calibration**: Recover target-city decay parameters $(\hat{\beta}, \hat{\gamma})$ via MLE on the target city's aggregate distance-bin histogram (the type of population-level summary routinely available from mobile network operators; see Appendix A). In the absence of target-city aggregate data, a national prior $(\beta_{\text{nat}}, \gamma_{\text{nat}})$ derived from the pooled source-city histograms may be used as a fallback.
2. **Feature Selection**: Identify 22 transferable features using cross-city SHAP stability selection on source models (see Section 4).
3. **Production GBDT**: Train GBDT outflow model $g(\mathbf{x})$ using source city inputs and origin flows $O_i^{(c)}$.
4. **Attraction Heuristic**: Define the training-free attraction function $a(\mathbf{x}_j) = \frac{1}{2}\left(\text{minmax}(P_j) + \text{minmax}(\text{POI}_j)\right)$.
5. **Zero-Shot Target Prediction**:
   a. Predict target city outflows: $\hat{O}_i^{(t)} = g(\mathbf{x}_i^{(t)})$.
   b. Compute target city attractions: $\hat{A}_j^{(t)} = a(\mathbf{x}_j^{(t)})$.
   c. Compute adaptive target offset: $\delta^{(t)} = 0.5 \times \bar{R}_{\text{zone}}^{(t)}$.
   d. Fuse components: $\hat{T}_{ij}^{(t)} = \hat{O}_i^{(t)}(d_{ij} + \delta^{(t)})^{-\hat{\gamma}} e^{-\hat{\beta} d_{ij}} \hat{A}_j^{(t)}$.

Table 2 Data usage matrix illustrating the operational roles of inputs across SDF-ZTF components to guarantee leakage-free prediction.

Data Source	**Outflow O_i**	**Attraction A_j**	**Distance Decay $f(d)$**	**Evaluation**
OD Matrix (T_{ij})	×	×	✓	✓
Aggregate Trip-Length Distribution	×	×	✓	×
Distance Matrix (d_{ij})	×	×	✓	✓
OpenStreetMap (OSM)	✓	✓	×	×
WorldPop	✓	✓	×	×

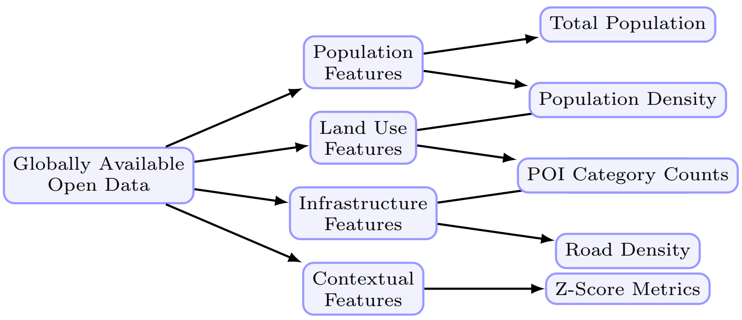

Fig. 4 Taxonomy of the globally available open data features used in the SDF-ZTF framework. Features are categorized into four dimensions: population, land use, infrastructure, and contextual scale factors.

scale differences, capturing the relative hierarchy of zones within their respective metropolitan areas.

4.3.2 SHAP Stability Selection

We select features using a cross-city SHAP stability score S_p. For each source city c, a local GBDT is trained and the mean absolute SHAP value $\mu_p^{(c)}$ is computed for feature p. The stability score is the signal-to-noise ratio across cities:

$$S_p = \frac{\mu_p}{\sigma_p + \epsilon} \tag{5}$$

where $\mu_p = \frac{1}{C}\sum_c \mu_p^{(c)}$ is the average feature importance, σ_p is the standard deviation of importances across cities, and $\epsilon = 10^{-5}$ prevents division by zero.

A feature p is retained in the final transfer set if: (1) its average importance is meaningful ($\mu_p \geq 0.005$), and (2) its importance is highly stable across different metropolitan layouts ($S_p \geq 1.0$, indicating that cross-city variance does not exceed the mean). This systematic procedure selects exactly 22 transferable features, achieving a 26.7% reduction in dimensionality while preserving zero-shot prediction accuracy.

4.4 Cross-City Transfer Evaluation Protocol

Standard mobility evaluations split OD pairs within a single city, which leaks spatial information from test zones into training. We instead use a strict cross-city protocol:

- **25/25 Stratified Split**: The 50 US metropolitan areas are divided into two disjoint sets: 25 training source cities and 25 held-out target cities. Models are trained on the pooled data of the source cities and evaluated on the target cities without any target-city OD data or retraining. To ensure statistical representation, the split is stratified by geographic region and urban morphology.
- **50-Fold Leave-One-City-Out (LOCO)**: The framework is iteratively trained on $C-1$ cities (49 cities) and evaluated on the single remaining held-out city, repeating this cycle for all 50 metropolitan areas to ensure robust out-of-distribution evaluation.

4.4.1 Information Leakage Prevention

Under this cross-city evaluation protocol, we enforce a strict information firewall between the source and target environments.

- **Forbidden**: Target-city flow matrices T_{ij}, target-city household survey results, target cellular data, and target trip-demand signals.
- **Allowed**: Centroid coordinates, administrative zone boundaries, OpenStreetMap POIs, and gridded open demography (WorldPop).

Since destination attraction is computed via a training-free open-data heuristic and distance decay parameters are learned on source cities, zero-shot leakage is mathematically prevented.

4.4.2 Evaluation Metric: Common Part of Commuters

We measure prediction quality with the Common Part of Commuters (CPC) (Lenormand et al. 2012):

$$\mathrm{CPC}(T, \hat{T}) = \frac{2\sum_{i,j}\min(T_{ij}, \hat{T}_{ij})}{\sum_{i,j} T_{ij} + \sum_{i,j} \hat{T}_{ij}} \tag{6}$$

where T_{ij} represents the ground-truth commuter flow from zone i to zone j, and $\hat{T}_{ij}$ is the predicted flow. The CPC index ranges from 0 (indicating completely disjoint flow matrices) to 1 (indicating perfect prediction).

4.5 Preprocessing & Normalization

Data preprocessing details are summarized in Table 3.

5 Results

5.1 Validation Framework Overview

We evaluate SDF-ZTF in three stages (Fig. 5): (1) validate each component individually, (2) quantify component contributions via factorial ablation, and (3) compare zero-shot transfer performance against baselines. All models are trained on 25 source cities and tested on 25 held-out targets with no target-city OD data.

5.2 Component Validation

We first validate each component in isolation.

5.2.1 Decay Recovery Validation

We compare two estimation modes for (γ, β) under a singly-constrained gravity model with fixed A_j (OSM heuristic) and ground-truth O_i:

- *Ground Truth (GT)*: Fitted directly on individual origin-destination (OD) pairs.
- *Recovered*: Fitted on the aggregate distance-bin distribution (marginal histogram of trip lengths, 50 percentile bins) using the aggregate MLE methodology.

Table 3 Preprocessing and normalization steps applied to spatial features prior to model learning.

Step	Treatment	Rationale
Missing Data	Median imputation	Rare ($< 2\%$); resolved from spatial neighbors
Log Transform	Population, POI counts	Stabilize right-skewed distributions
Standardization	Per-city (not global)	Preserve relative spatial hierarchy
Outliers	Retain	Important spatial hubs (CBDs); robust models

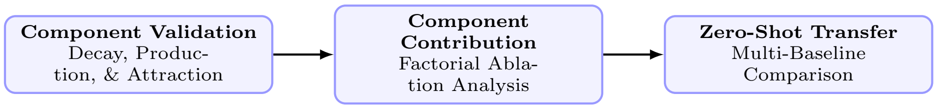

Fig. 5 Methodology validation roadmap showing the three evaluation stages used to systematically verify the SDF-ZTF framework.

Additional robustness sweeps are in Appendix A.4.

Pure Decay Recovery We fit the Tanner Multi decay function (Eq. 2) to the aggregate distance distribution of each city's training data. We compare the recovered parameters ($\hat{\gamma}_{\text{recovered}}, \hat{\beta}_{\text{recovered}}$) against the Ground Truth parameters ($\hat{\gamma}_{\text{gt}}, \hat{\beta}_{\text{gt}}$) fit directly on individual OD pairs.

Finding: As shown in Fig. 6, the parameter recovery shows reasonable correlation across the 25 target cities:

- **Power-law exponent** (γ): MAE $= 0.1313$, RMSE $= 0.1558$, Pearson correlation $r = 0.729$.
- **Exponential decay rate** (β): MAE $= 0.0111$, RMSE $= 0.0135$, Pearson correlation $r = 0.258$ (units in km^{-1}).

Recovery is reasonable for γ ($r = 0.729$). The weaker β recovery ($r = 0.258$) is expected given the noisy tail under aggregate binning, but the combined decay profile is accurate enough for OD prediction.

Note: Bin Sensitivity Analysis (sweeping $K \in \{10, 20, 50, 100\}$ distance bins) is detailed in Appendix A.4, validating that recovery accuracy stabilizes for $K \geq 50$.

Additional robustness experiments (including sensitivity to distance bin configurations, missing attractiveness data, and CBD collapse scenarios) and the corresponding recovery error heatmap are detailed in Appendix A.4.

5.2.2 Origin Production Transferability

To isolate the error from O_i estimation, we freeze A_j (OSM heuristic) and (γ, β) (national prior), then compare three outflow configurations on the 25 held-out cities: a naive population baseline ($O_i \propto \text{Population}_i$), GBDT-predicted outflows, and oracle ground-truth outflows. Results are in Table 4.

GBDT outflows (CPC 0.7037) outperform the naive baseline (0.584) and reach 94.5% of the oracle upper bound (0.7447). Robustness analyses are in Appendix B.

5.2.3 Transferable Feature Discovery

SHAP importance and stability scores are computed only on the 25 source cities (no target leakage). Table 5 lists the top-10 features.

Population and activity-related POIs are the most stable transferable predictors.

SHAP Feature Count Sweep. CPC plateaus at $K = 15$ features (0.7124) and stays flat through $K = 22$. We select $K = 22$ to retain all features meeting both thresholds ($\mu_p \geq 0.005$, $S_p \geq 1.0$), providing a stability margin for unseen cities. See Fig. 7.

5.3 Component Contribution Analysis

We perform a factorial ablation across the 25 held-out cities, replacing each disabled component with uniform weighting.

Table 6 shows that distance decay $f(d)$ is by far the most important: removing it drops CPC by

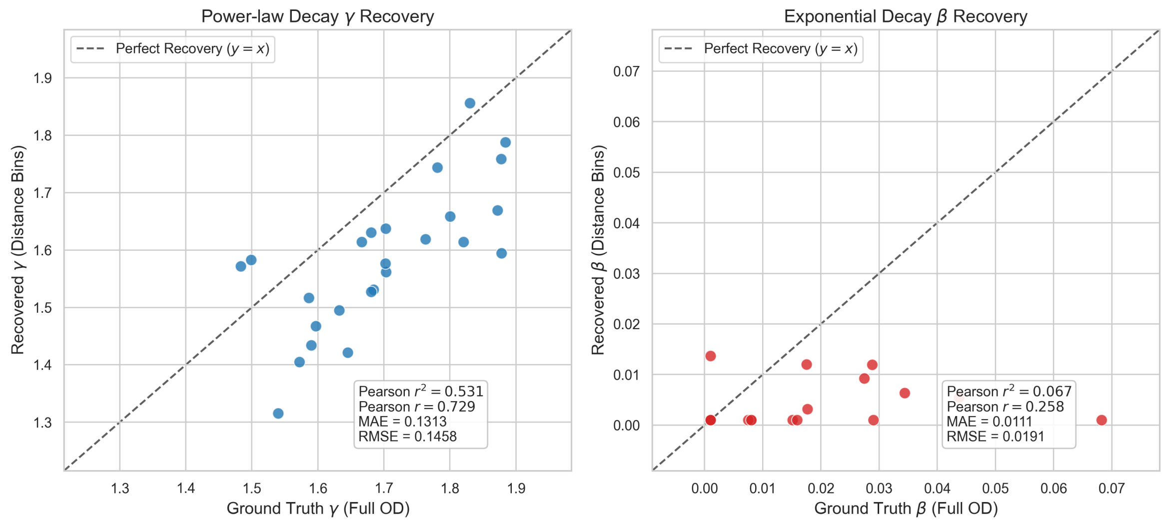

Fig. 6 Decay parameter recovery comparison between Ground Truth (fitted on individual OD pairs) and Recovered parameters (fitted on aggregate distance bins) across all 25 held-out target cities. Power-law exponent γ (left) and exponential decay rate β (right) show consistency in spatial decay structure (Pearson $r = 0.729$ for γ and $r = 0.258$ for β).

Table 4 Comparative Outflow Performance (Average CPC across $N = 25$ held-out cities)

Configuration	Mean CPC
Naive Population Baseline	0.584
Predicted Outflow (GBDT)	0.7037
Oracle Outflow (O_i^{true})	0.7447

31.1% (75.7% of explainable variance). Removing A_j and O_i causes 5.3% and 4.7% drops respectively (12.9% and 11.4% of variance). The ranking $f(d) \gg A_j > O_i$ is consistent across all 25 cities, confirming that all three components contribute but distance decay dominates.

Note on Baseline CPC: The baseline CPC of 0.7040 in this ablation analysis is obtained using the fixed national decay prior ($\gamma_{\text{nat}}, \beta_{\text{nat}}$) to freeze the decay component and isolate relative contributions under a uniform setting. When deploying the full SDF-ZTF (Survey-Free) framework with city-specific recovered decay parameters ($\hat{\gamma}, \hat{\beta}$), the performance increases to 0.7124 CPC (as reported in Table 7).

5.4 Zero-Shot Transfer Evaluation

We now evaluate the full SDF-ZTF pipeline against baselines.

5.4.1 Baseline Comparison

All models are trained on 25 source cities and tested zero-shot on 25 held-out targets. Table 7 summarizes results.

SDF-ZTF (Survey-Free) matches DeepGravity (CPC 0.7124 vs. 0.7117), and SDF-ZTF (Oracle O_i) reaches 0.7447. The two locally-calibrated Tanner baselines both yield CPC 0.750, serving as a structural ceiling; SDF-ZTF (Oracle O_i) approaches this ceiling without any target-city data.

5.4.2 Statistical Significance Analysis

Table 8 summarizes paired tests across all 50 LOCO folds. DeepGravity holds a small but significant edge under LOCO ($\Delta\text{CPC} = 0.0255, p < 10^{-10}$), but this gap disappears under the 25/25 split.

SDF-ZTF is statistically equivalent to Traditional Gravity ($p > 0.3$) and strongly outperforms

Table 5 Top-10 Features by Average SHAP Importance (Training Source Cities, $N = 25$)

Rank	Feature	μ_p	S_p
1	total_population	0.232	7.12
2	poi_food_fast	0.084	8.28
3	total_pois	0.057	5.33
4	z_total_pois	0.056	6.45
5	road_density_alt	0.053	2.21
6	road_density	0.050	2.39
7	poi_education_primary	0.030	5.57
8	area_km2	0.025	5.45
9	poi_shopping_supermarket	0.020	8.34
10	poi_entertainment_culture	0.018	3.55

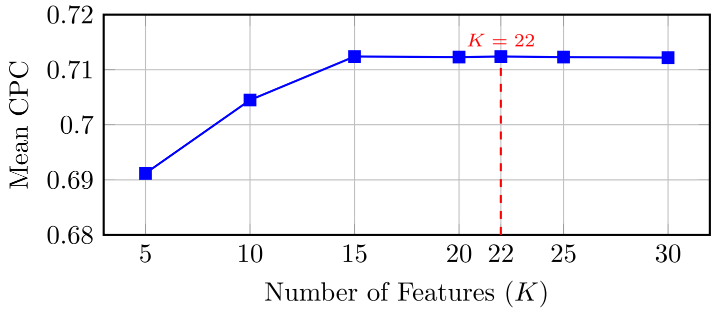

Fig. 7 SHAP feature count performance sweep showing zero-shot with 22 features retains stability

the Radiation Model ($p < 10^{-15}$, Cohen's $d = 6.28$).

5.4.3 Urban Morphology Analysis

The overall zero-shot performance masks spatial variations across urban configurations. We evaluate transfer CPC by urban morphology class across the 25 held-out cities in Table 9.

SDF-ZTF performs best in coastal and polycentric cities (CPC 0.734 and 0.716), while DeepGravity has a slight edge in dense transit environments (0.695 vs. 0.690). This suggests gravity decomposition works well where distance decay dominates, while neural models handle complex multi-modal transit better.

6 Discussion

6.1 What Aggregate Distributions Reveal About Distance Decay

The results in Section 5 answer the three research questions from Section 1. Aggregate trip-length histograms calibrate the decay function well ($r = 0.729$ for γ; Fig. 6). Distance decay accounts for 75.7% of explainable variance (Table 6). SDF-ZTF achieves CPC of 0.7124 under zero-shot conditions, matching DeepGravity (0.7117) without any target-city OD data (Table 7). In short, a compact set of spatial factors—distance friction, population density, and POI activity—can match the predictive capacity of neural models under cross-city transfer.

6.2 Distance Decay as the Dominant Transferable Mechanism

The 75.7% variance share of $f(d)$ reflects the physical constraint that distance imposes on commuting. While A_j and O_i determine where people want to go and how many trips originate, distance sets the feasible boundaries. Decay rates vary across urban layouts (e.g., coastal vs. grid cities), making this the hardest component to transfer. Our results show that aggregate trip-length histograms are sufficient to recover city-specific decay parameters, bypassing the need for individual OD trajectories and enabling privacy-preserving calibration.

Table 6 Factorial Ablation Results (Average across $N = 25$ held-out target cities)

$\mathbf{O}_i$	f(d)	$\mathbf{A}_j$	CPC	Contribution
✓	✓	✓	0.7040	Baseline (all components)
×	✓	✓	0.671	O_i removal: 4.7% loss
✓	×	✓	0.485	f(d) removal: 31.1% loss
✓	✓	×	0.666	A_j removal: 5.3% loss

Table 7 Multi-Baseline Comparison (25 Held-Out Target Cities)

Model	Mean CPC	Std Dev	Win % (vs DG)	Type	Features
SDF-ZTF (Survey-Free)	0.7124	0.0329	48.0%	Decomposed	22 (selected)
SDF-ZTF (Oracle O_i)	0.7447	0.028	56.0%	Decomposed	22 (selected)
DeepGravity (E2E)	0.7117	0.0332	–	Neural E2E	30 (all)
Tanner Gravity (Power Const.)	0.750	0.026	72.0%	Structure-only	0
Traditional Tanner Gravity	0.750	0.027	72.0%	Structure-only	0
Traditional Exponential Gravity	0.670	0.035	24.0%	Structure-only	0
Radiation Model	0.512	0.051	0.0%	Param-free	0

6.3 Positioning Within Existing Literature

Compared with classical gravity models, SDF-ZTF replaces local calibration with a GBDT-based production model and a training-free attraction heuristic. Compared with neural models like DeepGravity, it achieves comparable CPC (0.7124 vs. 0.7117) while remaining interpretable. Compared with existing transfer-learning frameworks, it requires zero target-city OD data.

6.4 Practical Implications

The decomposed structure offers practical advantages for planning:

Policy Simulation : Planners can modify A_j or O_i independently for "what-if" analyses without confounding the distance decay behavior.
Auditability : Each component is separately auditable, which supports regulatory approval and public accountability.
Survey-Free Deployment : The framework works in data-scarce regions by recovering decay from aggregate histograms and estimating attractions from open data.

6.5 Limitations

1. **Separability Assumption**: The model assumes approximate separability between production, attraction, and decay. In cities with highly integrated multi-modal transit networks or strong spatial interaction effects, this independence assumption may be violated, leading to localized prediction errors.
2. **Geographic Scope**: The evaluation is limited to US metropolitan areas. Cities in other regions (e.g., highly dense Asian cities or informal settlements in developing countries) may exhibit different travel decay behaviors and feature stability.
3. **Temporal Dynamics**: The model uses a pre-pandemic baseline (2019) and does not capture changes in travel friction induced by remote work patterns or modal shifts.
4. **Demographic Representation**: Input datasets like cellular location logs underrepresent elderly, low-income, and smartphone-free populations, introducing potential demographic bias in the output OD flows (Gallotti et al. 2024).
5. **OSM coverage**: Urban areas are comprehensively covered, but the rural fringe has sparse POI data. Transferability to low-density zones remains uncertain.

Table 8 LOCO 50-Fold Statistical Significance Test Summary

Baseline Model	Mean Diff.	t-stat (t)	p-value (t-test)	Wilcoxon W	p-value (Wilc.)	95% CI Diff.	Cohen's d
DeepGravity	-0.0255	-9.150	3.53×10^{-12}	66.0	3.49×10^{-10}	[-0.0311, -0.0199]	-1.29
Traditional Gravity	0.0000	1.000	3.22×10^{-1}	18.5	3.31×10^{-1}	[-0.0000, 0.0000]	0.14
Radiation Model	0.2492	44.416	3.01×10^{-41}	0.0	1.78×10^{-15}	[0.2380, 0.2605]	6.28

Table 9 Zero-Shot Performance by Urban Morphology (25 Held-Out Target Cities)

Morphology Type	# Cities	SDF-ZTF CPC	DeepGravity CPC	Gravity Win %
Dense / Transit	5	0.690	0.695	20%
Sprawling / Grid	11	0.715	0.718	45%
Polycentric	6	0.716	0.712	50%
Coastal	3	0.734	0.712	66%

6. **Causality**: Analysis is purely predictive. Causal effects cannot be inferred (e.g., building a grocery store attracts X more trips). Causal inference requires experimental or quasi-experimental design.

7 Conclusions

This study addressed three questions about transferable mobility structure through zero-shot evaluation across 50 US metropolitan areas.

First, aggregate trip-length histograms are sufficient to calibrate distance decay. The recovered γ correlates well with OD-fitted parameters ($r = 0.729$ across 25 held-out cities), confirming that anonymized distribution summaries serve as an effective calibration signal.

Second, distance decay dominates: it accounts for 75.7% of explainable variance, with removal causing a 31.1% CPC drop—far exceeding the losses from removing attraction (5.3%) or production (4.7%).

Third, the decomposed framework matches neural baselines. Using 22 SHAP-selected open-data features, SDF-ZTF achieves zero-shot CPC of 0.7124, on par with DeepGravity (0.7117) under the same protocol.

These results indicate that transferable mobility structure is more regular than commonly assumed. Identifying stable structural mechanisms from open data offers a transparent and globally deployable alternative to data-intensive neural approaches.

Appendix: Supplementary Materials

Appendix A. Robustness of Aggregate-Based Distance-Decay Recovery

A.1 Estimation Details

The distance decay parameters (γ, β) are estimated via maximum likelihood estimation (MLE) from the aggregate trip-length distribution (marginal distance histogram) of source cities. To ensure numerical stability and enforce parameter bounds ($\beta > 0, \gamma \geq 0$), we perform L-BFGS-B optimization in log-space, parameterizing the variables as $\tilde{\beta} = \log(\beta)$ and $\tilde{\gamma} = \log(\gamma)$. The optimization employs a multi-restart framework with 10 random perturbations initialized around baseline data-driven anchors to prevent the solver from stalling in local optima, retaining the parameter set that maximizes log-likelihood.

A.2 Robustness Analysis

We evaluate the robustness of our MLE parameter recovery under non-ideal data scenarios:

Bin Sensitivity Analysis Sweeping the distance bins $K \in \{10, 20, 50, 100\}$ shows that the recovery error for the power-law exponent γ stabilizes at a low level (MAE ≈ 0.13) for $K \geq 50$. This indicates that 50 percentile bins are sufficient to robustly capture the decay profile.

Missing Attractiveness Data (A_j) Randomly removing 10%, 20%, and 50% of target-city destination attractions A_j yields highly stable decay recovery (e.g., $\gamma_{\mathrm{MAE}} = 0.1312$ under 20% missing data vs. 0.1313 baseline). This validates the robustness of aggregate MLE against localized data gaps.

CBD Collapse (Extreme A_j Distortion) Setting the attraction A_j of the top 5% highest attractiveness zones to zero still allows accurate parameter recovery ($\gamma_{\mathrm{MAE}} = 0.1225$). This proves that spatial decay friction can be recovered even when key destination hubs are omitted.

Appendix B. Robustness of Origin Production Transfer

The origin trip production (O_i) estimator is trained using a Gradient Boosted Decision Tree (GBDT) regressor on source training cities. Complete hyperparameter configurations and implementation scripts are available in the project's code repository. Here, we analyze the sensitivity of the zero-shot production transfer.

B.1 Source-Scale Sensitivity

We analyze how the number of training environments affects zero-shot predictability by retraining the GBDT on subsets of $n_{\mathrm{src}} \in \{5, 10, 15, 20, 25\}$ source cities. Target-city evaluation shows that zero-shot prediction performance rises steeply from $n_{\mathrm{src}} = 5$ and saturates around 15–20 source cities. This indicates that only a modest number of source cities is required to learn stable, generalizable outflow characteristics.

B.2 Noise Sensitivity

To evaluate the stability of origin transfer under prediction errors, we inject additive Gaussian noise at log-scale ($\sigma \in \{10\%, 20\%, 40\%\}$) directly onto the predicted outflows $\log \hat{O}_i$. Even under substantial injected noise (40%), the target CPC degrades only marginally (from 0.7037 to 0.6923). This demonstrates that production-constraint normalization effectively dampens localized estimation noise.

Appendix C. Feature Selection Sensitivity

C.1 Feature Count Sensitivity

Evaluating the GBDT model performance across nested subsets of top SHAP features ($K \in \{5, 10, 15, 20, 22, 25, 30\}$) shows that predictive performance plateaus between 15 and 22 features (CPC: 0.7124). This indicates that most transferable spatial information is captured by a compact set of built-environment indicators, with additional features providing marginal utility.

A.6 List of Abbreviations

The key abbreviations and acronyms used throughout this paper are summarized and defined in Table 11.

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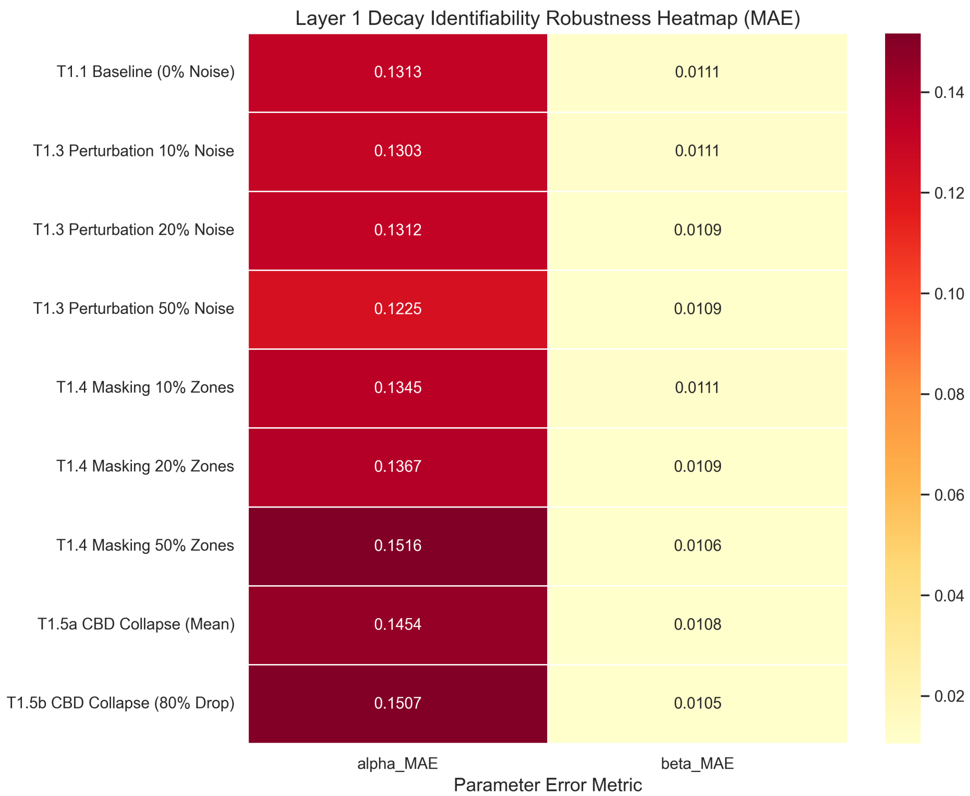

Fig. 8 Robustness of Aggregate-Based Distance-Decay Recovery (MAE) under varying levels of noise, missing data, and CBD collapse scenarios across all 25 held-out target cities.

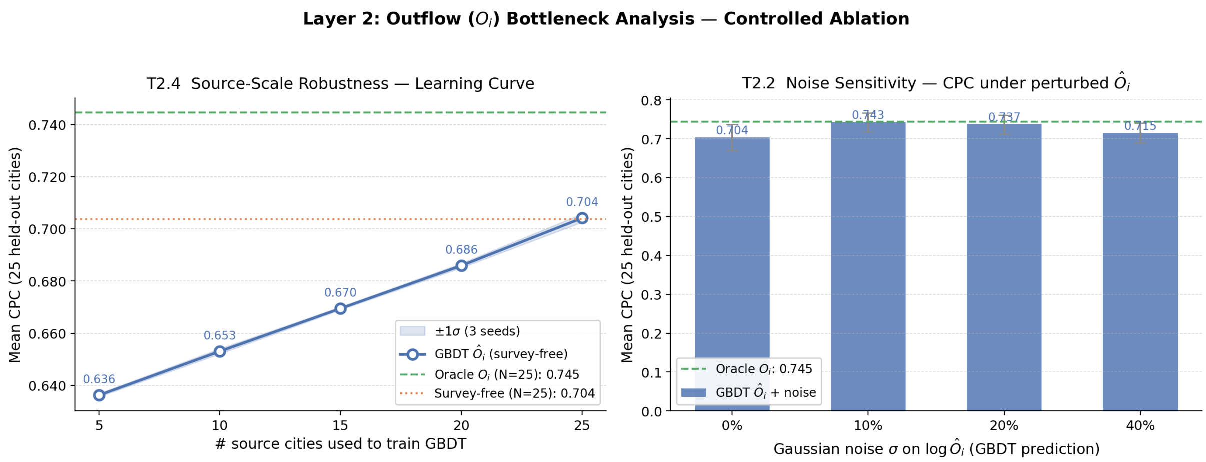

Fig. 9 Outflow bottleneck analysis. **Left**: CPC vs. training source cities (n_{src}), showing performance saturation at 15–20 cities. **Right**: Noise sensitivity of CPC under additive Gaussian noise σ injected on predicted outflows $\log \hat{O}_i$.

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Table 10 GBDT Model Hyperparameter Configurations

Hyperparameter	Outflow Model (O_i)	Attraction Model (A_j, ablation only)
Base Learner	Decision Tree	Decision Tree
Loss Function	Least Squares (MSE on $\log O_i$)	Least Squares (MSE on A_j)
Learning Rate (η)	0.05	0.05
Number of Estimators	150	200
Max Tree Depth	6	6
Num. Leaves	31	31
Subsample Ratio	0.8	0.8
Colsample By Tree	0.8	0.8
Minimum Child Weight	1	1
Early Stopping Rounds	15	15

Table 11 List of Abbreviations

Abbreviation	Definition	Abbreviation	Definition
CBD	Central Business District	MSE	Mean Squared Error
CPC	Common Part of Commuters	OD	Origin-Destination
GBDT	Gradient Boosted Decision Tree	OSM	OpenStreetMap
GDPR	General Data Protection Regulation	POI	Point of Interest
LOCO	Leave-One-City-Out	SDF-ZTF	Separable Decomposed Framework for Zero-Shot Transf
MLE	Maximum Likelihood Estimation	SHAP	SHapley Additive exPlanations

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